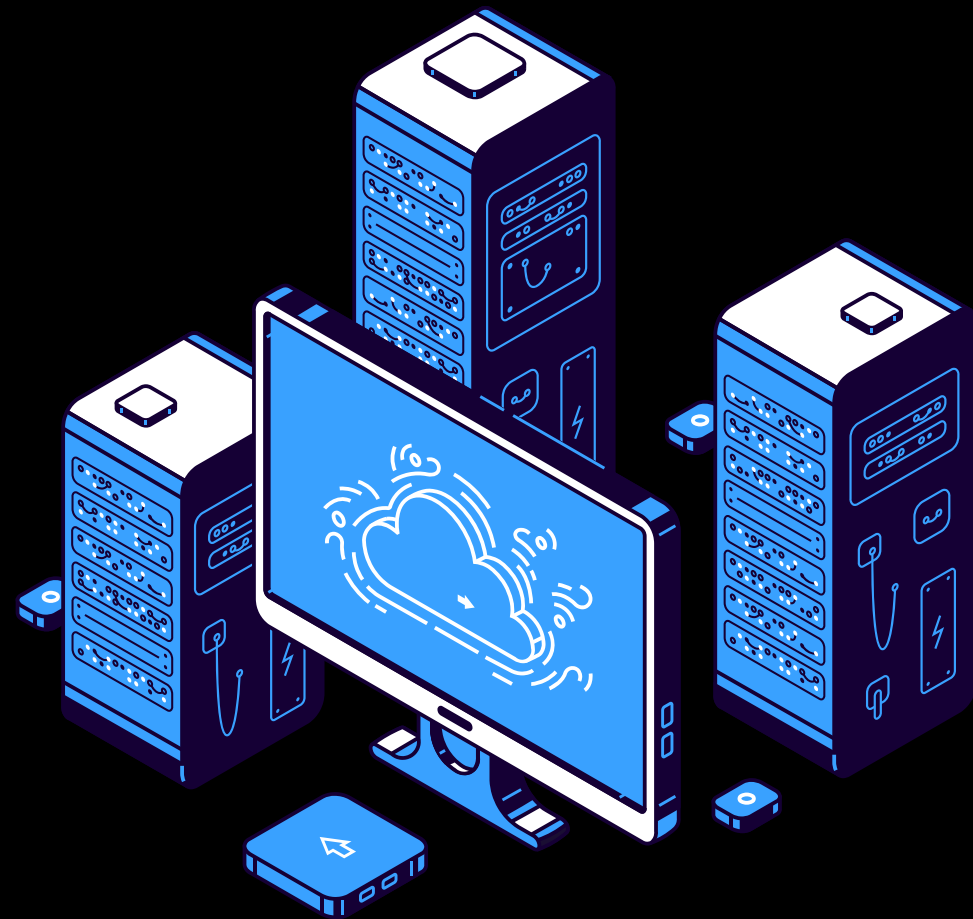


LAPORAN OSPF

Computers process information efficiently, enabling communication, analysis, and automation. This deck fundamentals, components, software, networks, security, applications, and directions.

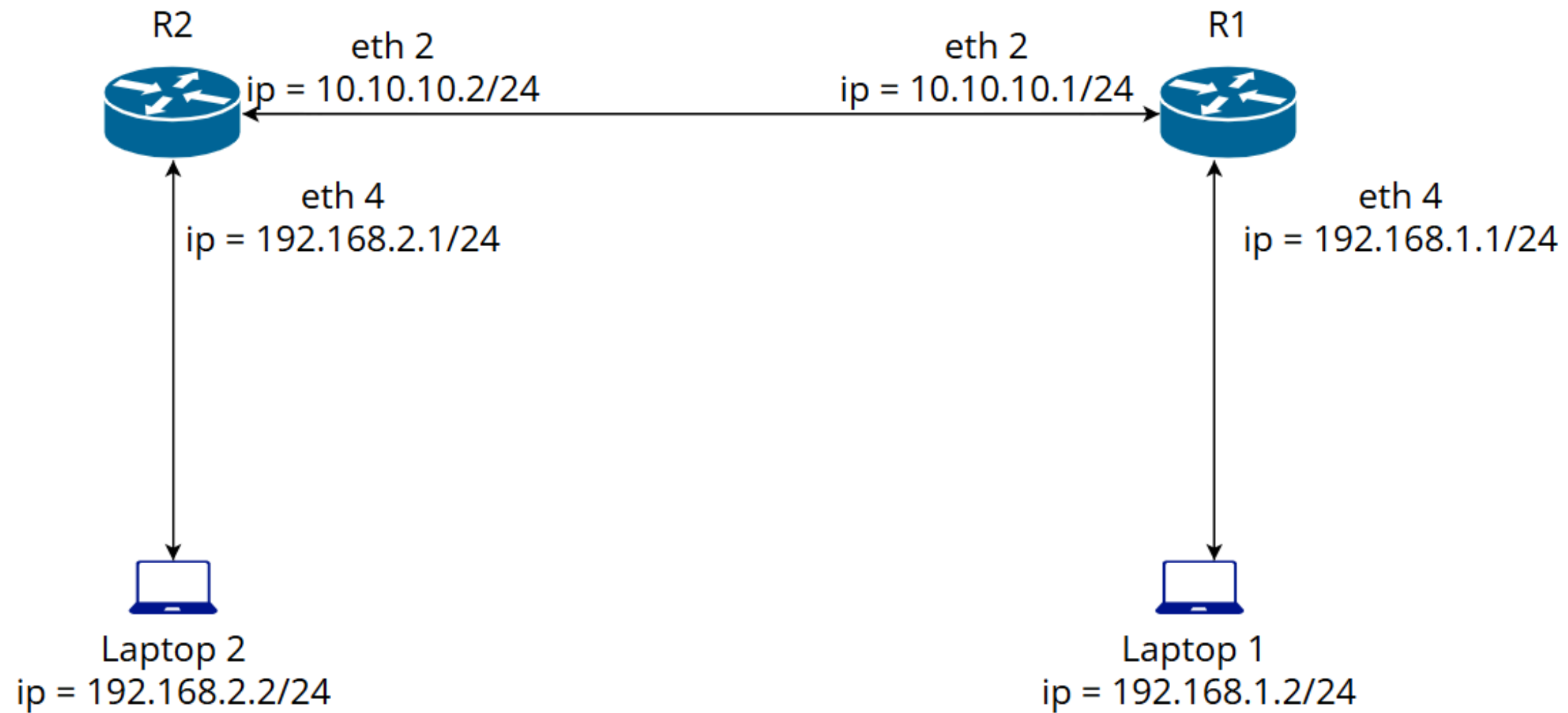


ALAT DAN BAHAN



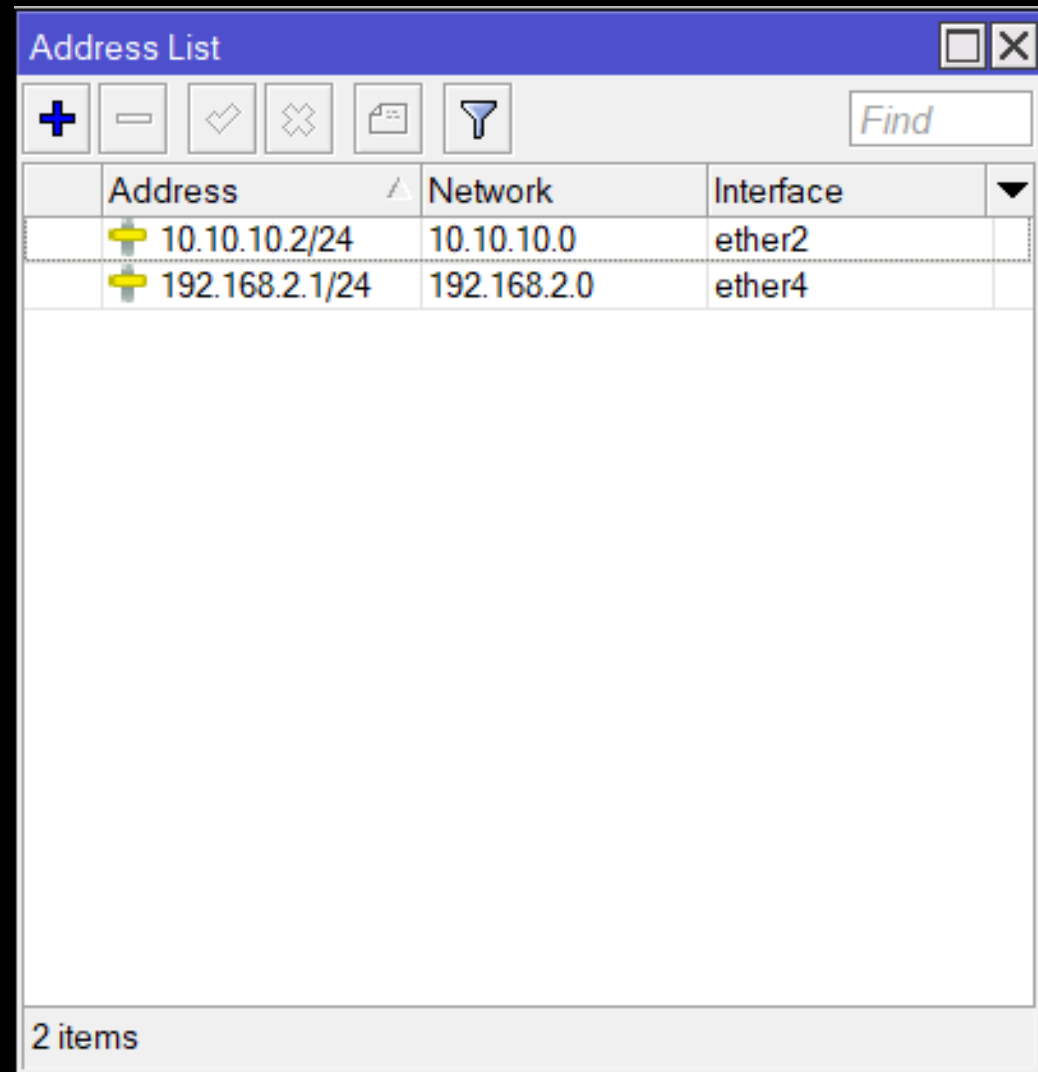
- 3 UNIT KABEL (LAN)
- 2 UNIT LAPTOP
- 2 UNIT ROUTER

TOPOLOGI



KONFIGURASI IP ADDRESS PADA ROUTER

1 DAN 2



Address	Network	Interface
10.10.10.2/24	10.10.10.0	ether2
192.168.2.1/24	192.168.2.0	ether4

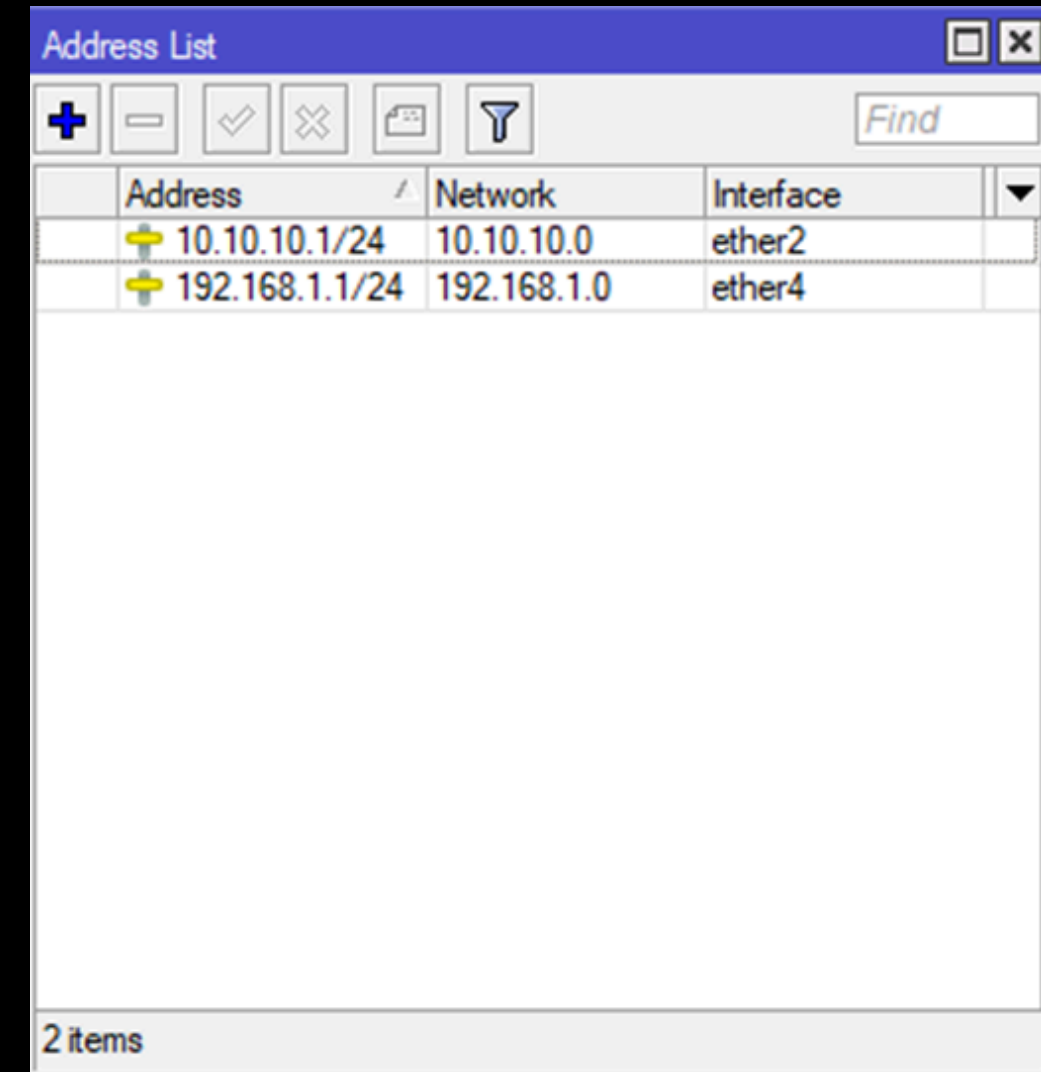
2 items

ROUTER 2 :

- Address = 10.10.10.2/24
- Network = 10.10.10.0
- Interface = ether 2

ROUTER 2 :

- Address = 192.168.2.1/24
- Network = 192.168.2.0
- Interface = ether 4



Address	Network	Interface
10.10.10.1/24	10.10.10.0	ether2
192.168.1.1/24	192.168.1.0	ether4

2 items

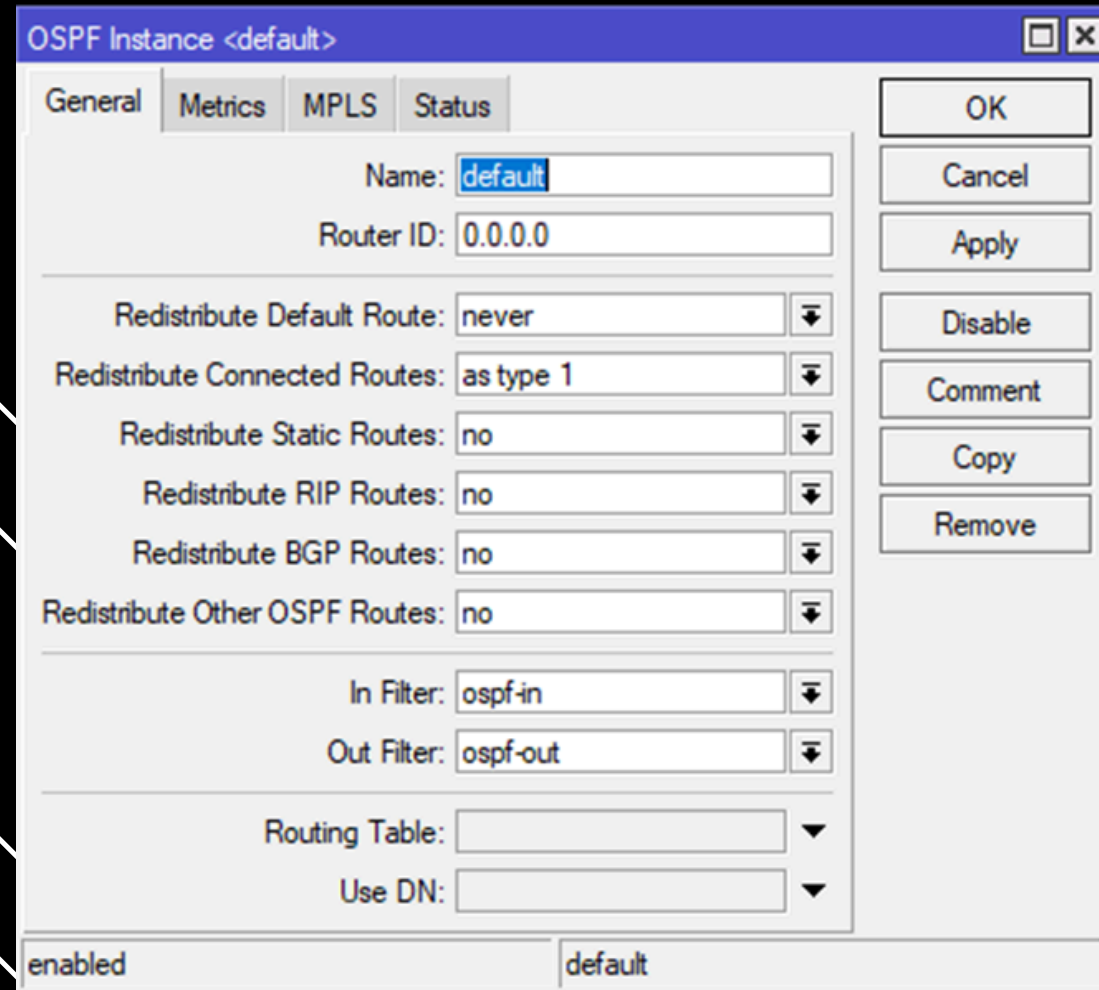
ROUTER 1 :

- Address = 10.10.10.1/24
- Network = 10.10.10.0
- Interface = ether2

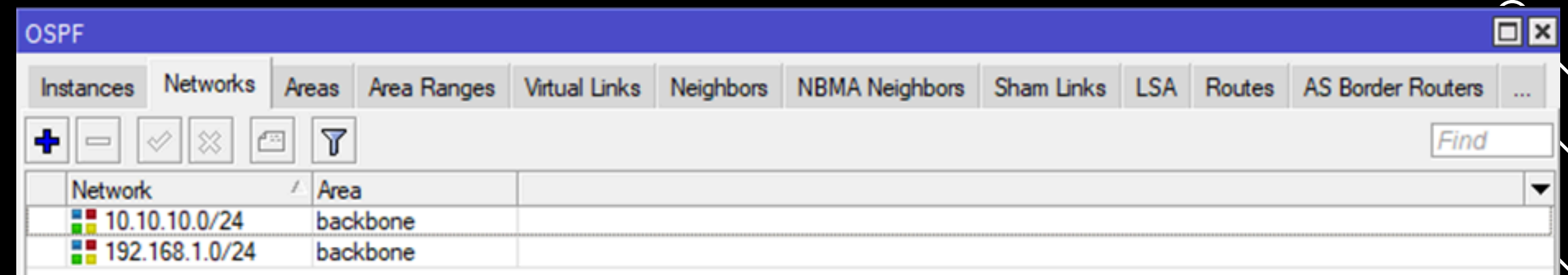
ROUTER 1 :

- Address = 192.168.1.1/24
- Network = 192.168.1.0
- Interface = ether 4

KONFIGURASI DYNAMIC ROUTING OSPF DAN IP LAPTOP PADA ROUTER 1

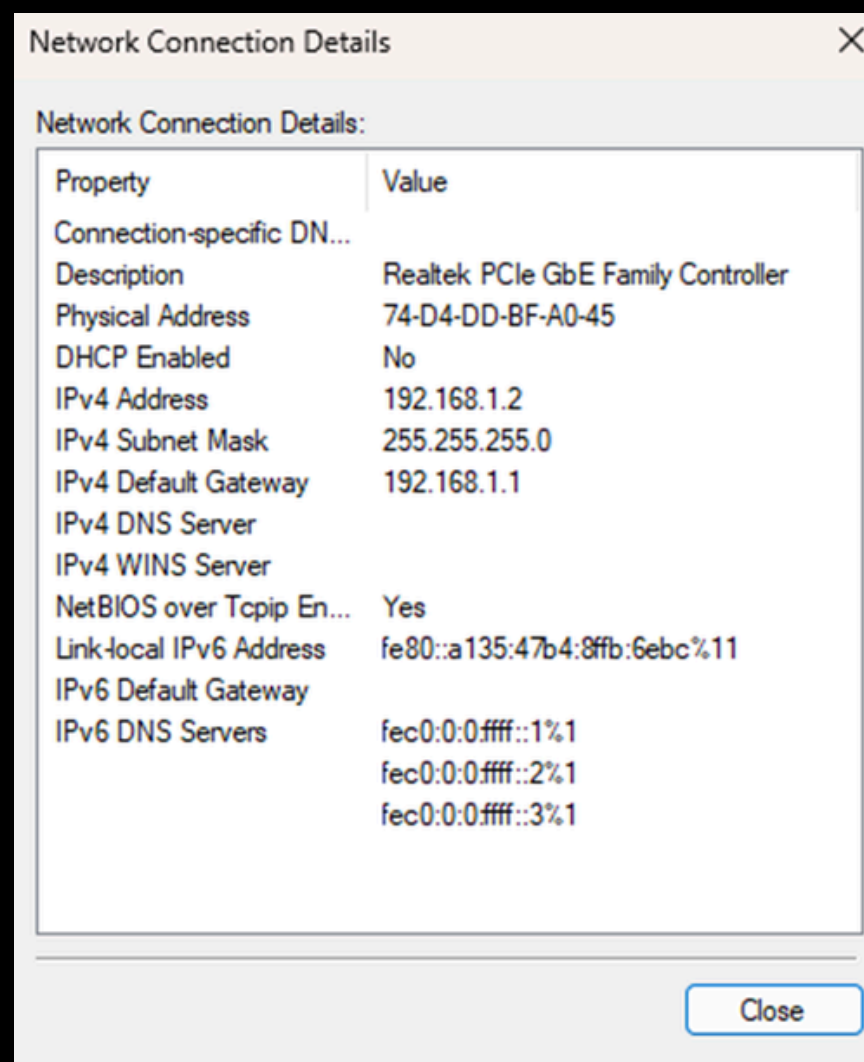


- Pilih tab Instances> klik default
- Pada pengaturan OSPF Instances, di kolom "Resdritibute Connected Routes"
- pilih "as type 1"



- Pilih tab Networks > klik icon +
- Network = 192.168.1.0/24
- Area = backbone
- Masih pada tab Networks > klik icon +
- Network = 10.10.10.0/24
- Area = backbone

KONFIGURASI DYNAMIC ROUTING OSPF DAN IP LAPTOP PADA ROUTER 1



- Klik open network & Internet setting
- Pilih Ethernet
- Pilih change adapter option
- Pilih ethernet
- Double klik Internet Protocol Version 4 (TCP/IPv4)
- Pilih use the following IP address :
 - - IP Address = 192.168.1.2
 - - Subnetmask = 255.255.255.0
 - - Default gateway = 192.168.1.1

KONFIGURASI DYNAMIC ROUTING OSPF DAN IP LAPTOP PADA ROUTER 2

OSPF Instance <default>

General Metrics MPLS Status

Name: default

Router ID: 0.0.0.0

Redistribute Default Route: never

Redistribute Connected Routes: as type 1

Redistribute Static Routes: no

Redistribute RIP Routes: no

Redistribute BGP Routes: no

Redistribute Other OSPF Routes: no

In Filter: ospf-in

Out Filter: ospf-out

Routing Table:

Use DN:

OK Cancel Apply Disable Comment Copy Remove

enabled default

- Klik menu Routing > OSPF
- Pilih tab Instances> klik default
- Pada pengaturan OSPF Instances, di kolom "Resdritibute Connected Routes"
- pilih "as type 1"

OSPF

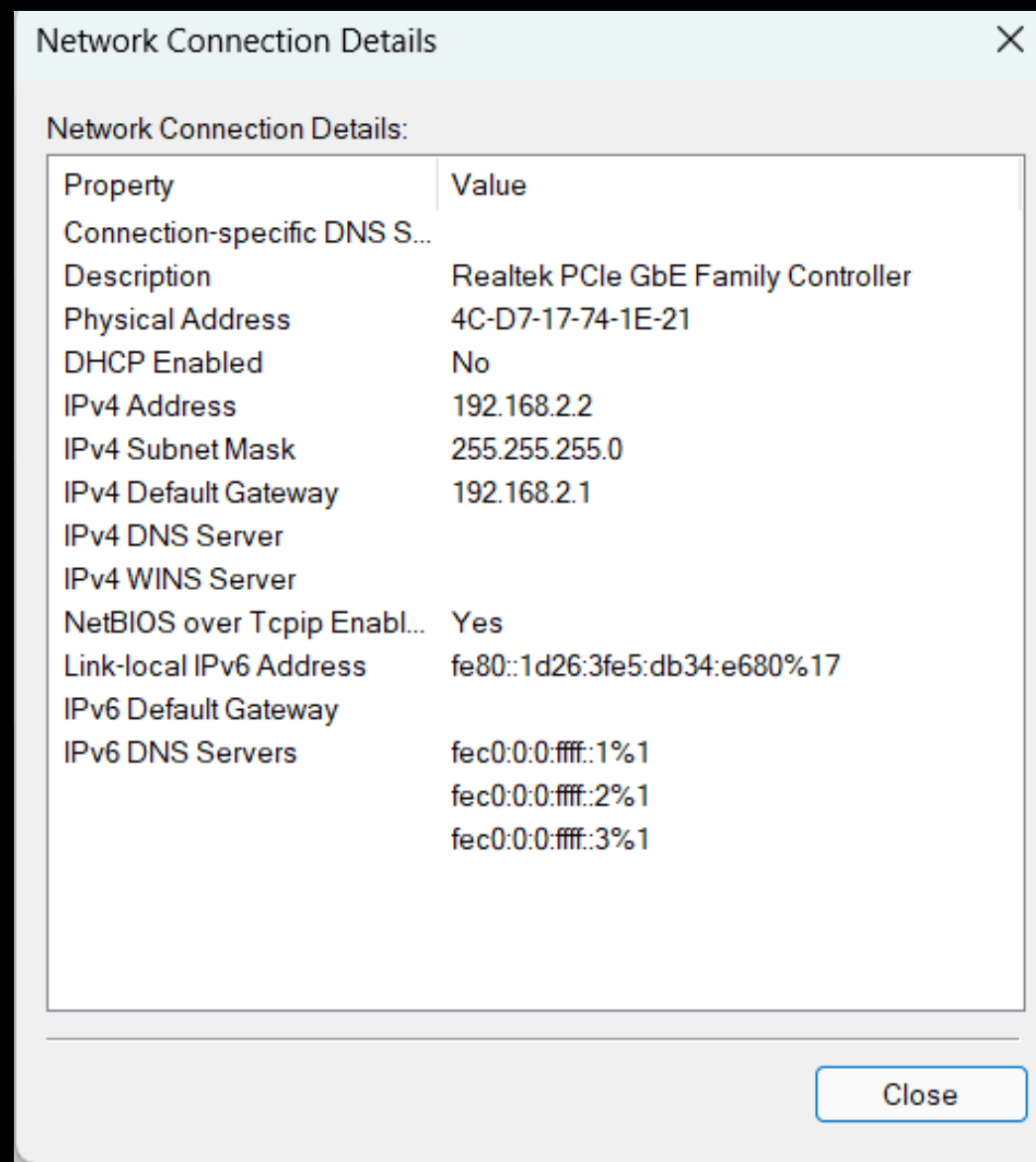
Instances Networks Areas Area Ranges Virtual Links Neighbors NBMA Neighbors Sham Links LSA Routes AS Border Routers ...

+ - ✓ ✗ [icon] [icon] Find

Network	Area
10.10.10.0/24	backbone
192.168.2.0/24	backbone

- Pilih tab Networks > klik icon +
- Network = 192.168.2.0/24
- Area = backbone
- Masih pada tab Networks > klik icon +
- Network = 10.10.10.0/24
- Area = backbone

KONFIGURASI DYNAMIC ROUTING OSPF DAN IP LAPTOP PADA ROUTER 2



- Klik open network & Internet setting
- Pilih Ethernet
- Pilih change adapter option
- Pilih ethernet
- Double klik Internet Protocol Version 4 (TCP/IPv4)
- Pilih use the following IP address
- - IP Address = 192.168.2.2
- - Subnetmask = 255.255.255.0
- - Default gateway = 192.168.2.1

PENGUJIAN

LAPTOP 1

```
C:\Windows\System32>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Windows\System32>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=3ms TTL=126
Reply from 192.168.2.2: bytes=32 time=2ms TTL=126
Reply from 192.168.2.2: bytes=32 time=3ms TTL=126
Reply from 192.168.2.2: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Windows\System32>tracert 192.168.2.2

Tracing route to DESKTOP-R6EE5AL [192.168.2.2]
over a maximum of 30 hops:

  0  1 ms    2 ms    1 ms   192.168.1.1
  1  2 ms    2 ms    1 ms   10.10.10.2
  2  3 ms    2 ms    2 ms   DESKTOP-R6EE5AL [192.168.2.2]

Trace complete.

C:\Windows\System32>
```

LAPTOP 2

```
C:\Windows\System32>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time=1ms TTL=64
Reply from 192.168.2.1: bytes=32 time=1ms TTL=64
Reply from 192.168.2.1: bytes=32 time=1ms TTL=64
Reply from 192.168.2.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Windows\System32>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=3ms TTL=126
Reply from 192.168.1.2: bytes=32 time=3ms TTL=126
Reply from 192.168.1.2: bytes=32 time=4ms TTL=126
Reply from 192.168.1.2: bytes=32 time=4ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\Windows\System32>tracert 192.168.1.2

Tracing route to LAPTOP-3AG1MMLO [192.168.1.2]
over a maximum of 30 hops:

  0  1 ms    <1 ms   <1 ms   192.168.2.1
  1  1 ms    <1 ms   <1 ms   10.10.10.1
  2  3 ms    3 ms    2 ms    LAPTOP-3AG1MMLO [192.168.1.2]

Trace complete.
```

THANK YOU